

Punto 5 Filtros de Blueprints:

1. Implementa filtros:
- ****RedundancyFilter****: elimina puntos duplicados consecutivos.

```
/**
 * Elimina puntos consecutivos duplicados (x,y) para reducir redundancia.
 * Perfil: "redundancy"
 */
@Component
@Profile("redundancy")
public class RedundancyFilter implements BlueprintsFilter {
    @Override
    public Blueprint apply(Blueprint bp) {
        List<Point> in = bp.getPoints();
        if (in.isEmpty()) return bp;
        List<Point> out = new ArrayList<>();
        Point prev = null;
        for (Point p : in) {
            if (prev == null || !(prev.x()==p.x() && prev.y()==p.y())) {
                out.add(p);
                prev = p;
            }
        }
        return new Blueprint(bp.getAuthor(), bp.getName(), out);
    }
}
```

- ****UndersamplingFilter****: conserva 1 de cada 2 puntos.

```
/**
 * Undersampling: conserva 1 de cada 2 puntos (índices pares), reduciendo la densidad.
 * Perfil: "undersampling"
 */
@Component
@Profile("undersampling")
public class UndersamplingFilter implements BlueprintsFilter {
    @Override
    public Blueprint apply(Blueprint bp) {
        List<Point> in = bp.getPoints();
        if (in.size() <= 2) return bp;
        List<Point> out = new ArrayList<>();
        for (int i = 0; i < in.size(); i++) {
            if (i % 2 == 0) out.add(in.get(i));
        }
        return new Blueprint(bp.getAuthor(), bp.getName(), out);
    }
}
```

- Agregamos @Profile al IdentityFilter para que solo esté activo un perfil cuando **no** se usa ningún otro perfil.

```
/**
 *
 */
@Component
@Profile("!redundancy & !undersampling")
public class IdentityFilter implements BlueprintsFilter {
    @Override
    public Blueprint apply(Blueprint bp) { return bp; }
}
```

2. Activa los filtros mediante perfiles de Spring (`redundancy`, `undersampling`).

Proceso para Redundancy

- i. Levantamos la app con el perfil de redundancy no sin antes crear las tablas para probarlo correctamente en Swagger.

```
CREATE TABLE
CREATE TABLE
blueprintsbd=# \q
PS C:\Users\robin\Downloads\LAB04_ARSW> mvn spring-boot:run "-Dspring-boot.run.profiles=redundancy"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.eci.arsw:SpringBoot_REST_API_Blueprints >-----
[INFO] Building SpringBoot_REST_API_Blueprints 1.0.0
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] >>> spring-boot:3.3.9:run (default-cli) > test-compile @ SpringBoot_REST_API_Blueprints >>>
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ SpringBoot_REST_API_Blueprints ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ SpringBoot_REST_API_Blueprints ---
[INFO] Nothing to compile - all classes are up to date.

[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ SpringBoot_REST_API_Blueprints ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] <<< spring-boot:3.3.9:run (default-cli) < test-compile @ SpringBoot_REST_API_Blueprints <<<
[INFO] <<< spring-boot:3.3.9:run (default-cli) < test-compile @ SpringBoot_REST_API_Blueprints <<<
[INFO] <<< spring-boot:3.3.9:run (default-cli) < test-compile @ SpringBoot_REST_API_Blueprints <<<
[INFO]
[INFO] --- spring-boot:3.3.9:run (default-cli) @ SpringBoot_REST_API_Blueprints ---
[INFO] Attaching agents: []

:: Spring Boot :: (v3.3.9)

2026-02-19T23:04:56.236-05:00 INFO 26076 --- [main] e.e.a.blueprints.BlueprintsApplication : Starting BlueprintsApplication using Java 21.0.8 with PID 26076 (C:\Users\robin\Do
target\classes started by robin in C:\Users\robin\Downloads\LAB04_ARSW)
2026-02-19T23:04:56.240-05:00 INFO 26076 --- [main] e.e.a.blueprints.BlueprintsApplication : The following 1 profile is active: "redundancy"
2026-02-19T23:04:58.333-05:00 INFO 26076 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-02-19T23:04:58.520-05:00 INFO 26076 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-02-19T23:04:58.520-05:00 INFO 26076 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.36]
2026-02-19T23:04:58.611-05:00 INFO 26076 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-02-19T23:04:58.613-05:00 INFO 26076 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 2302 ms
2026-02-19T23:04:59.777-05:00 INFO 26076 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-02-19T23:04:59.797-05:00 INFO 26076 --- [main] e.e.a.blueprints.BlueprintsApplication : Started BlueprintsApplication in 4.00 seconds (process running for 4.665)
```

- ii. Luego en Swagger mediante el método de POST creamos un blueprint con puntos duplicados.

```
Curl

curl -X 'POST' \
  'http://localhost:8080/api/v1/blueprints' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "author": "test",
    "name": "filtro",
    "points": [
      {"x":1,"y":1},
      {"x":1,"y":1},
      {"x":2,"y":2},
      {"x":3,"y":3},
      {"x":3,"y":3}
    ]
  }'
```

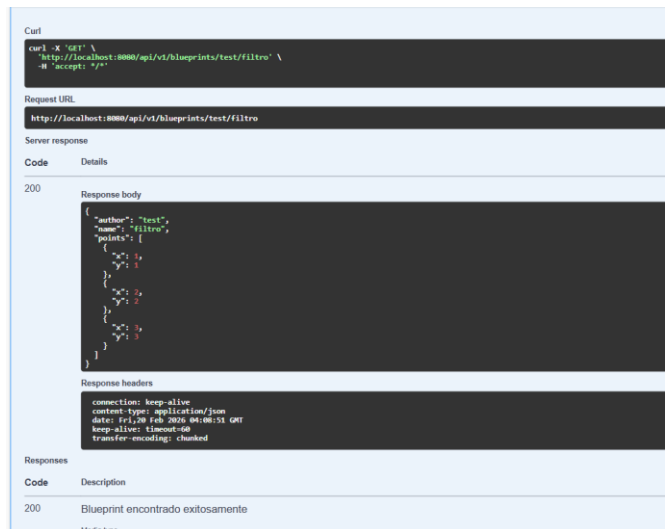
Request URL

```
http://localhost:8080/api/v1/blueprints
```

Server response

Code	Details
201	Response headers <pre>connection: keep-alive content-length: 0 date: Fri, 20 Feb 2026 04:07:16 GMT keep-alive: timeout=60</pre>

- iii. Consultamos con GET /api/v1/blueprints/test/filtro y se puede observar solo 3 puntos sin los duplicados demostrando que el filtro funciono correctamente. (1,1), (1,1), (2,2), (3,3), (3,3) => (1,1), (2,2), (3,3).



Proceso para Undersampling

- i. Levantamos con el nuevo perfil Undersampling

```
¿Desea terminar el trabajo por lotes (S/N)? S
PS C:\Users\robin\Downloads\LAB04_ARSM> mvn spring-boot:run "-Dspring-boot.run.profiles=undersampling"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.eci.arsw:SpringBoot_REST_API_Blueprints >-----
[INFO] Building SpringBoot_REST_API_Blueprints 1.0.0
[INFO]    from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:3.3.9:run (default-cli) > test-compile @ SpringBoot_REST_API_Blueprints >>>
[INFO]
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 0 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.13.0:compile (default-compile) @ SpringBoot_REST_API_Blueprints ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ SpringBoot_REST_API_Blueprints ---
[INFO] skip non existing resourceDirectory C:\Users\robin\Downloads\LAB04_ARSM\src\test\resources
[INFO]
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ SpringBoot_REST_API_Blueprints ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:3.3.9:run (default-cli) < test-compile @ SpringBoot_REST_API_Blueprints <<<
[INFO]
[INFO] --- spring-boot:3.3.9:run (default-cli) @ SpringBoot_REST_API_Blueprints ---
[INFO] Attaching agents: []
```

```

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:: Spring Boot ::      (v3.3.9)

2026-02-19T23:22:33.575-05:00 INFO 22908 --- [main] e.e.a.blueprints.BlueprintsApplication : Starting BlueprintsApplication using Java 21.0.8 with PID 22908 (C:\Users\robin\O
2026-02-19T23:22:33.578-05:00 INFO 22908 --- [main] e.e.a.blueprints.BlueprintsApplication : The following 1 profile is active: "undersampling"
2026-02-19T23:22:35.395-05:00 INFO 22908 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080 (http)
2026-02-19T23:22:35.417-05:00 INFO 22908 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-02-19T23:22:35.418-05:00 INFO 22908 --- [main] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/10.1.36]
2026-02-19T23:22:35.522-05:00 INFO 22908 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-02-19T23:22:35.524-05:00 INFO 22908 --- [main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 1875 ms
2026-02-19T23:22:36.273-05:00 INFO 22908 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http) with context path '/'
2026-02-19T23:22:36.281-05:00 INFO 22908 --- [main] e.e.a.blueprints.BlueprintsApplication : Started BlueprintsApplication in 3.695 seconds (process running for 4.244)

```

ii. Creamos un nuevo blueprint con varios puntos

Curl

```
curl -X 'POST' \
  'http://localhost:8080/api/v1/blueprints' \
  -H 'accept: */*' \
  -H 'Content-Type: application/json' \
  -d '{
    "author": "juanComunica",
    "name": "filtro2",
    "points": [
      {"x":1,"y":1},
      {"x":2,"y":2},
      {"x":3,"y":3},
      {"x":4,"y":4},
      {"x":5,"y":5}
    ]
  }'
```

Request URL

http://localhost:8080/api/v1/blueprints

Server response

Code	Details
201	Response headers <pre>connection: keep-alive content-length: 0 date: Fri, 20 Feb 2026 04:25:53 GMT keep-alive: timeout=60</pre>

Responses

Code	Description
201	Blueprint creado exitosamente

- iii. Consultamos con GET /api/v1/blueprints/juanComunica/filtro2 y la respuesta es correcta ya que nos devuelve 3 puntos (1,1), (3,3), (5,5) que corresponden a los índices 0, 2, 4.

```
curl -X 'GET' \
'http://localhost:8080/api/v1/blueprints/juanComunica/filtro2' \
-H 'accept: */*'

```

Request URL

```
http://localhost:8080/api/v1/blueprints/juanComunica/filtro2

```

Server response

Code	Details
200	Response body <pre>{ "author": "juanComunica", "name": "filtro2", "points": [{ "x": 1, "y": 1 }, { "x": 3, "y": 3 }, { "x": 5, "y": 5 }] }</pre> Response headers <pre>connection: keep-alive content-type: application/json date: Fri, 20 Feb 2026 04:27:26 GMT keep-alive: timeout=60 transfer-encoding: chunked </pre>

Responses

Code	Description
200	Blueprint encontrado exitosamente